



Atmospheric Thermodynamics



Chapter 4: Water and Its Transformations



Objective

- *To Describe The Thermodynamic Properties Of Water*
- *To Apply The Van Der Waals Equation To Describe Real Gas*
- *To Explain Phase Transformations And Latent Heat*
- *To Use The Clausius–Clapeyron Equation To Quantify Saturation Vapor Pressure Changes*

Water and Its Transformations

Homogeneous, Closed Systems

- Only one phase (e.g., just gas or just liquid)
- No mass enters or leaves → closed system
- Thermodynamics here is simple: only 2 variables (e.g., T and p) needed

Heterogeneous Systems (More Realistic)

- Involve multiple phases (e.g., vapor + liquid water).
- Phases can exchange mass → like water evaporating or condensing.
- Treated as open subsystems within a closed total system

Dry Air vs. Water Components

- Dry air (N_2 , O_2 , CO_2 , Ar) is always gaseous → treated as closed.
- Water can be vapor, liquid, or ice → treated as open (changes phase)

Cont. . .

- Atmosphere Is a Mixed System
- Real atmosphere = dry air + water substance in various phases.
- Must model each component carefully to account for energy and mass transfer.

Thermodynamic Properties Of Water

Thermodynamic Properties Of Water

Property	Key Control	Notes
Vapor pressure	Temperature	Measures escaping tendency
Saturation vapor pressure	Temperature	Max vapor that air can hold
Van der Waals behavior	Pressure & volume	Needed for real gas correction

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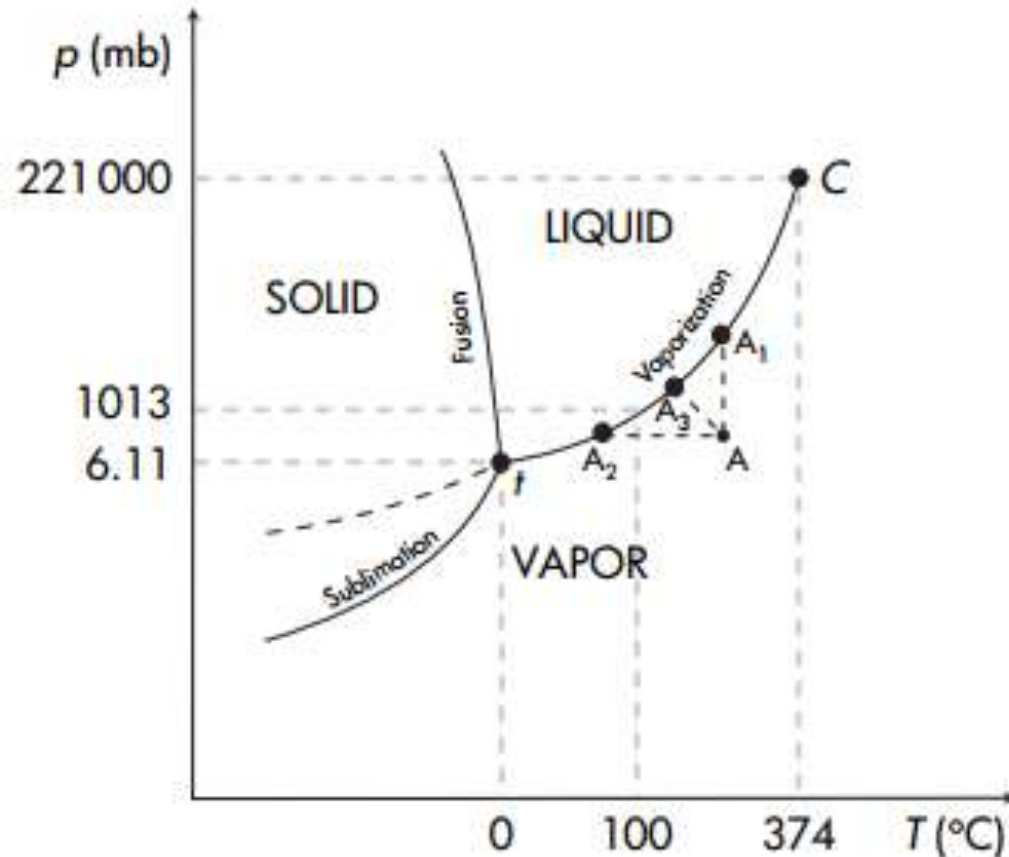
- These properties describe how water behaves under changes in temperature, pressure, and phase (solid, liquid, vapor)
- They relate to the energy and state of the system

Cont. . .

- Ideal vs. Non-Ideal Behavior
 - Water vapor alone behaves like an ideal gas, but when it coexists with liquid water or ice, the mixture no longer follows the ideal gas laws due to interactions between phases

- **Phase Transition Curves:** The phase diagram (Fig.) shows curves representing vaporization, sublimation, and fusion. Each curve shows where two phases coexist at equilibrium, and the pressure on these curves is called the saturation vapor pressure

Figure
Phase-transition equilibria
for water.



Vapor Pressure (e)

- The pressure exerted by water vapor in equilibrium with liquid water at a given temperature.
- **Increases with temperature.**
- Units: Pa or hPa
- Important for **evaporation and condensation** processes

Saturation Vapor Pressure (e_s)

- The maximum vapor pressure air can hold at a given temperature before condensation occurs.
- Describes the threshold for saturation.
- Over water and ice, the values differ due to phase differences.
- Commonly described by the Clausius–Clapeyron equation
- Saturation vapor pressure is smaller for ice than for water.

The Van der Waals Equation

What is the Van der Waals Equation

- The Van der Waals equation is a modified version of the ideal gas law that accounts for the non-ideal behavior of real gases.
- It's especially useful for describing gases like water vapor near condensation, where molecular interactions are significant.

Why is it a Thermodynamic Concept?

- It Describes how pressure, volume, and temperature interact in real gases (like water vapor).
- It Improves predictions of phase transitions, which are governed by energy and molecular interactions (core thermodynamics)

Cont. . .

- Helps evaluate enthalpy, internal energy, and work done during transformations.
- Water vapor, especially near condensation, deviates from ideal gas behavior.
- The Van der Waals equation helps capture that real gas behavior, making it a useful thermodynamic model for water in the atmosphere.

Cont. . .

Ideal Gas Law

$$PV = nRT$$

Assumes no molecular volume or attraction

Works best at high T, low P

Van der Waals Equation

$$\left(P + a\frac{n^2}{V^2}\right)(V - nb) = nRT$$

Adjusts for real behavior of molecules

Works better near condensation/saturation

Equilibrium phase transformations – latent heat

Process	Description	Heat
Evaporation	Liquid $\rightarrow$ vapor	absorbs heat
Condensation	Vapor $\rightarrow$ liquid	releases heat
Freezing	Liquid $\rightarrow$ ice	releases heat
Melting	Ice $\rightarrow$ liquid	absorbs heat
Sublimation	Ice $\rightarrow$ vapor	absorbs heat

Latent Heat (L)

- Heat required for phase change at constant pressure without changing temperature.
- Latent heat is the change in enthalpy (ΔH) during the phase change

Latent Heat Types

L_v : Latent heat of vaporization (liquid $\leftrightarrow$ vapor).

L_f : Latent heat of fusion (ice $\leftrightarrow$ liquid).

L_s : Latent heat of sublimation (ice $\leftrightarrow$ vapor)

Positive latent heat: Heat is absorbed (e.g., liquid $\rightarrow$ vapor, ice $\rightarrow$ liquid).

Negative latent heat: Heat is released (e.g., vapor $\rightarrow$ liquid, liquid $\rightarrow$ ice)

Latent Heat Equations:

- $L_v = H_v - H_w = U_v - U_w + p_{wv}(V_v - V_w).$
- $L_f = H_w - H_i = U_w - U_i + p_{wi}(V_w - V_i).$
- $L_s = H_v - H_i = U_v - U_i + p_{vi}(V_v - V_i).$

Triple Point:

At the triple point, the equilibrium pressure between two phases is $p = 6.11 \text{ mb}$.

At the triple point: $l_s = l_f + l_v$.

The Clausius–Clapeyron Equation

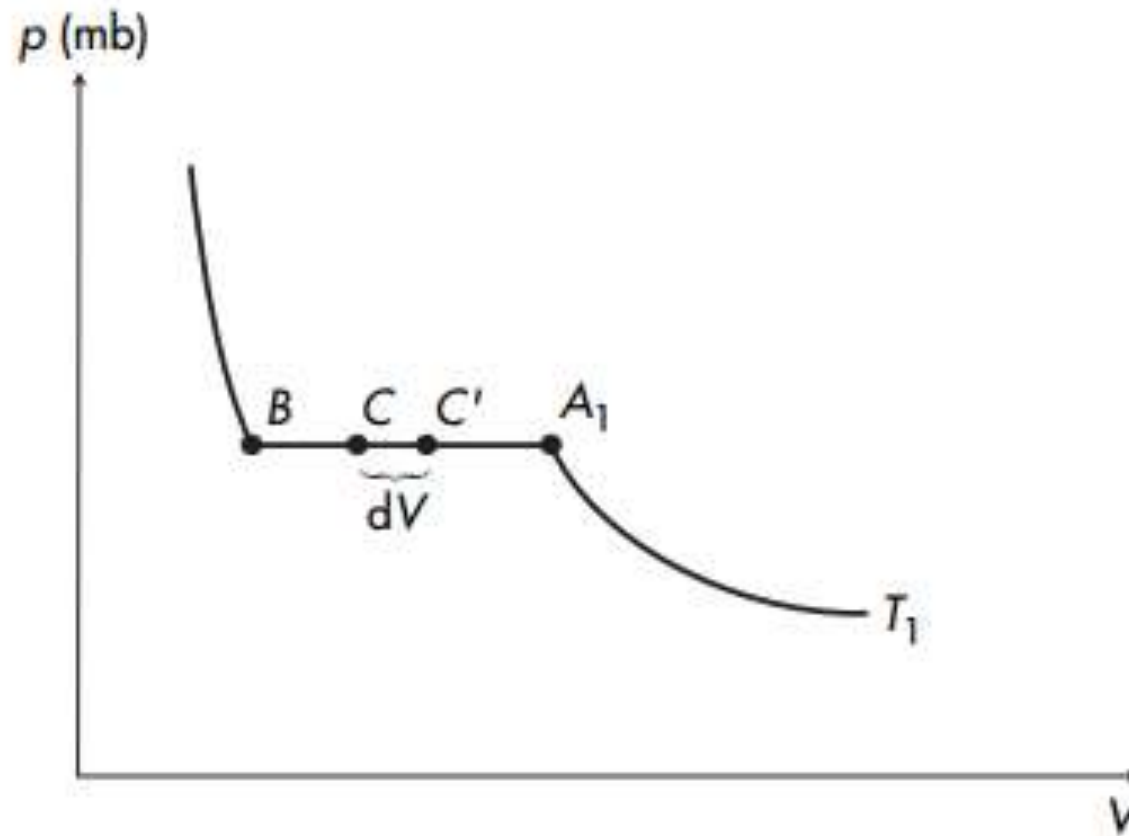
Clausius–Clapeyron Equation

- It describes how saturation vapor pressure changes with temperature during a phase change (e.g., liquid $\leftrightarrow$ vapor, solid $\leftrightarrow$ vapor)

Cont. . .

Figure

The isotherm T_1 of Figure 6.2. For any given change $C \rightarrow C'$ the change in volume is dV . This causes an amount dm of liquid water to evaporate.



Cont. . .

$$\frac{de_s}{dT} = \frac{L_v}{T(\alpha_2 - \alpha_1)} = \frac{L_v}{T\Delta\alpha}$$

is called the Clausius–Clapeyron (C–C) equation.

- Here l_v is the specific latent heat (or specific enthalpy) of the change of phase
- e_s equilibrium vapor pressure
- $\Delta\alpha$ is the difference in the specific volumes between the two phases at temperature T

Cont. . .

- This equation shows that the rate of change of vapor pressure with temperature depends on:
 - The latent heat
 - The temperature
 - The volume difference between vapor and liquid

Why is it a Thermodynamic Equation?

- Because it comes from:
 - The first law of thermodynamics
 - The concept of equilibrium between phases
 - And uses key thermodynamic quantities: entropy, enthalpy, latent heat, volume

Cont. . .

- It connects thermodynamic properties like:
 - Latent heat (L)
 - Temperature (T)
 - Volume change (Δv) across a phase boundary
 - Vapor pressure (e_s)

This Makes It Fundamental In Meteorology For:

- Cloud formation
- Boiling/condensation levels
- Dew and frost prediction
- Saturation processes

Approximations Of The C–C Equation

- Latent heat (l_v , l_s , l_f) is nearly constant over typical atmospheric temperatures (-20°C to 30°C).
- Specific heats (c_p) of water vapor, liquid water, and ice vary little with temperature—often treated as constants.
- Water vapor behaves like an ideal gas; liquid water and ice do not

Cont. . .

- Saturation vapor pressure depends exponentially on temperature can be approximated using simplified formulas.

Consequences of the C–C Equation

- Saturation vapor pressure increases rapidly with temperature, especially in warm regions
- Warm air can hold more water vapor, leading to more intense evaporation and precipitation in the tropics

Cont. . .

- Latent heat release during condensation fuels atmospheric motion and helps drive the general circulation.
- Melting point of ice decreases with pressure, explaining glacier movement around obstacles.
- Boiling point of water decreases with lower pressure, affecting boiling at high altitudes

Adiabatic Transformations

(a) Dry Adiabatic Process

- $\Gamma_d = 9.8^\circ\text{C}/\text{km}$
- No condensation
- No latent heat
- Rising dry air cools at this rate

(b) Moist Adiabatic Process

- $\Gamma_m \approx 4\text{--}7^\circ\text{C}/\text{km}$
- Condensation occurs
- Latent heat is released

Cooling is slower than dry air

Mixing Processes

(a) Vertical Mixing

- Caused by convection and turbulence

Transfers:

- heat
 - moisture
 - momentum
- **Example:**
Surface heating → rising air → cloud formation

(b) Horizontal Mixing (Advection)

- Movement of air masses horizontally
- Warm air advection or Cold air advection
- **Example:**

Warm moist air moving over cold surface → fog

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Thanks !

Atmospheric Thermodynamics

Chapter 5 : Measures And Description Of Moist Air

Objective

- *To describe the Humidity variables*
- *To know the relationship between humidity variables*
- *To Explain Processes in the atmosphere*

Humidity Variables

- Water vapor behaves like an ideal gas, and we can express its properties in terms of pressure, temperature, and mass
- We use different humidity variables (q , w , r) to describe moisture in the air, depending on what comparison we're making (to dry air, total air, or saturation)

Cont. . .

- The formulas can be simplified in most atmospheric applications since the actual amount of moisture is small relative to air mass
- The relative humidity shows how close the air is to being saturated with water vapor

Important ideas from chapter one

- Water vapor in moist air follows the

ideal gas law: $e = \rho_v R_v T$

e : vapor pressure

ρ_v : density of water vapor

R_v : gas constant for water vapor (≈ 461.5
 $\text{J/kg}\cdot\text{K}$)

T : temperature in Kelvin

Epsilon (ϵ or $\beta = 0.622$)

- This is the ratio of the molar masses of water vapor and dry air:

- $\beta = M_v / M_d = R_d / R_v \approx 0.622$

- M_v : molar mass of water = 18.01 g/mol
- M_d : molar mass of dry air ≈ 28.97 g/mol

Specific Humidity (q) Vs. Mixing Ratio (w)

- The mass of water vapor per total mass of moist air:

$$q = \rho_v / \rho = m_v / m$$

- The mass of water vapor per mass of dry air:

$$w = \rho_v / \rho_d = m_v / m_d$$

where $m = m_d + m_v$ is the total mass of the mixture (ρ_v is also called the absolute humidity). Since $\rho_v = eM_v/R^*T$, $\rho_d = p_d M_d / R^*T$, and $p_d = p - e$ it follows that

Cont. . .

$$w = \epsilon \frac{e}{p - e}.$$

At saturation the *saturation* mixing ratio is

$$w_s = \epsilon \frac{e_s}{p - e_s}$$

where e is either the equilibrium (saturation) vapor pressure with respect to liquid water (e_{sw}) or with respect to ice (e_{si}). In general $e_s, e \ll p$ which reduces the above two equations to

$$w \approx \epsilon \frac{e}{p}, \quad w_s \approx \epsilon \frac{e_s}{p}. \quad \text{--- (1)}$$

Cont. . .

Since $1/q = (m_d + m_v)/m_v = (m_d/m_v) + 1$ it follows that

$$\frac{1}{q} = \frac{1}{w} + 1$$

or

$$w = \frac{q}{1 - q}, \quad q = \frac{w}{1 + w}.$$

- In the atmosphere both w and q are very small ($w, q \ll 1$). For this reason we can always assume that $w \approx q$. The relative humidity, r , is defined as

Cont. . .

$$r = \frac{m_v}{m_{vs}}$$

- where m_v is the mass of water vapor in a sample of moist air of volume V and m_{vs} is the mass of water vapor the sample would have had if it were saturated. Because of the ideal gas law we can write r as

$$r = \frac{e}{e_s}.$$

Using (1) it follows that approximately

$$r \approx \frac{w}{w_s}.$$



2. Mean Molecular Weight of Moist Air

Dry air molecular weight:

$$M_d \approx 28.97$$

Water vapor:

$$M_v \approx 18$$



Moist air is lighter than dry air.

Virtual Temperature (T_v)

$$T_v = T(1 + 0.61q)$$



Moist air behaves like warmer dry air (less dense)

Processes in the atmosphere

Isobaric cooling – dew and frost temperatures

- Isobaric cooling refers to cooling at constant pressure.
- In a closed system, specific humidity q and mixing ratio w remain constant during cooling.
- Vapor pressure (e) stays constant until condensation occurs.
- Equilibrium vapor pressure (e_s) decreases as temperature drops, increasing relative humidity (RH).

Cont. . .

- When RH reaches 100%, the air becomes saturated:
 - If saturation occurs with respect to liquid water → dew point temperature (T_{dew}).
 - If with respect to ice → frost point temperature (T_f)

Adiabatic Isobaric Processes – Wet-Bulb Temperatures

- The wet-bulb temperature (T_w) is the lowest temperature that air can be cooled to by evaporating water into it, under constant pressure (isobaric) and without adding or removing heat from outside (adiabatic).
- A system of unsaturated moist air + liquid water tends to reach equilibrium by evaporating water.

Cont. . .

- The process is adiabatic (no heat exchange) and isenthalpic (constant enthalpy), causing temperature to decrease as water evaporates.
- The heat needed for evaporation is drawn from the air, resulting in cooling.

Cont. . .

- If there's sufficient liquid water, the system reaches saturation at the wet-bulb temperature (T_w).
- This process is spontaneous and irreversible.
- A real-world example: evaporative cooling from rain falling through unsaturated air.

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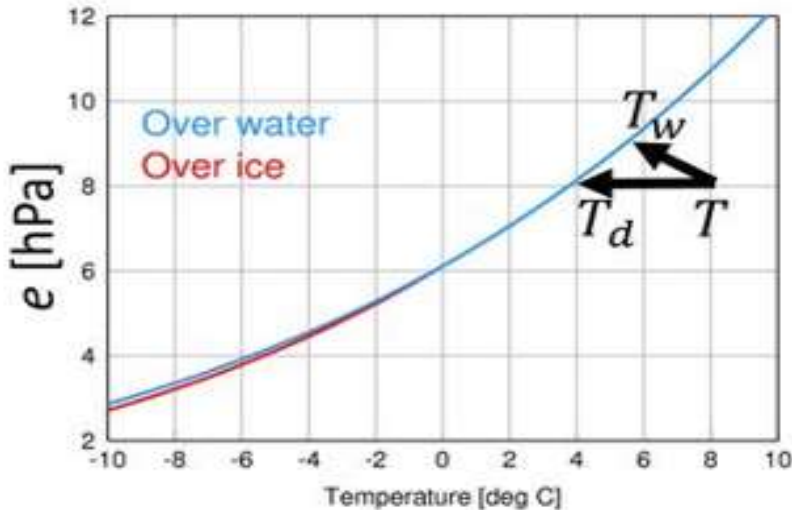
- If ice is involved instead of liquid water, the ice-bulb temperature (T_i) is defined.
- Wet-bulb temperature (T_w) $\neq$ Dew point temperature (T_{dew}) due to differing processes

🔍 Aspect	🌫️ Dew Point Temperature (T_{dew})	💧 Wet-Bulb Temperature (T_{w})
Definition	Temperature at which air becomes saturated when cooled at constant pressure (no moisture added)	Temperature reached when water evaporates into the air under adiabatic, isobaric conditions
Cooling Mechanism	Cooling without moisture change	Cooling due to evaporation (adds moisture)
Process Type	Reversible & based on condensation	Irreversible & based on evaporation
Atmospheric Use	Indicates actual moisture content	Used in wetting/cooling processes like evaporation of rain
Common Occurrence	Dew on surfaces during the night	Cooling effect felt when sweat evaporates
Relationship		

- Dew point temperature:

$$e_s(T_d) = e$$

“The temperature to which air must be cooled isobarically to achieve saturation”



- Wet bulb temperature:

T_w (get from *psychrometric tab*)

“The temperature to which air can be cooled isobarically by evaporating water into it”



Condensation and Sublimation Requirements

- For condensation or frost to form:
 - Condensation nuclei or surfaces must be present.
 - Otherwise, supersaturation is needed, which is rare in the atmosphere

Natural Processes Causing Isobaric Cooling

- Radiative cooling of surface/air → dew or frost forms.
- If air near the ground cools below dew point:
 - Dew forms if $T > 0^{\circ}\text{C}$
 - Frost forms if $T < 0^{\circ}\text{C}$
- Radiation fog forms when air layers cool below saturation.
- Advection fog forms when air moves over a colder surface and cools

Adiabatic Expansion (or Compression) of Unsaturated Moist Air

- Water vapor affects the properties of air Moist air behaves similarly to dry air in adiabatic processes, but the presence of water vapor can cause slight variations in temperature and potential temperature.
- Virtual temperature matters For atmospheric processes, considering the virtual temperature of moist air is crucial because it influences buoyancy and stability.

Cont. . .

- Lapse rates differ The presence of water vapor in moist air leads to a lower rate of temperature decrease with altitude compared to dry air.
- This is important for understanding atmospheric stability and convection.

Reaching saturation by adiabatic ascent

- Rising air parcel expands and cools adiabatically
- No condensation until saturation is reached
- Mixing ratio (w) and specific humidity (q) remain constant
- Vapor pressure (e) decreases with height
- Relative humidity ($r = e/e_s$) increases during ascent

Why Relative Humidity Increases During Ascent

- Pressure (p) decreases $\rightarrow$ e must decrease to keep e/p constant
- Equilibrium vapor pressure (e_s) decreases faster than e
- Hence, $r = e/e_s$ increases as long as e decreases slower than e_s

Cloud Formation and the LCL

- LCL (Lifting Condensation Level): Point where air becomes saturated ($r = 100\%$)
- TLCL: Temperature at LCL, where condensation starts
- Clouds form at and above LCL due to moisture condensation

Saturated ascent

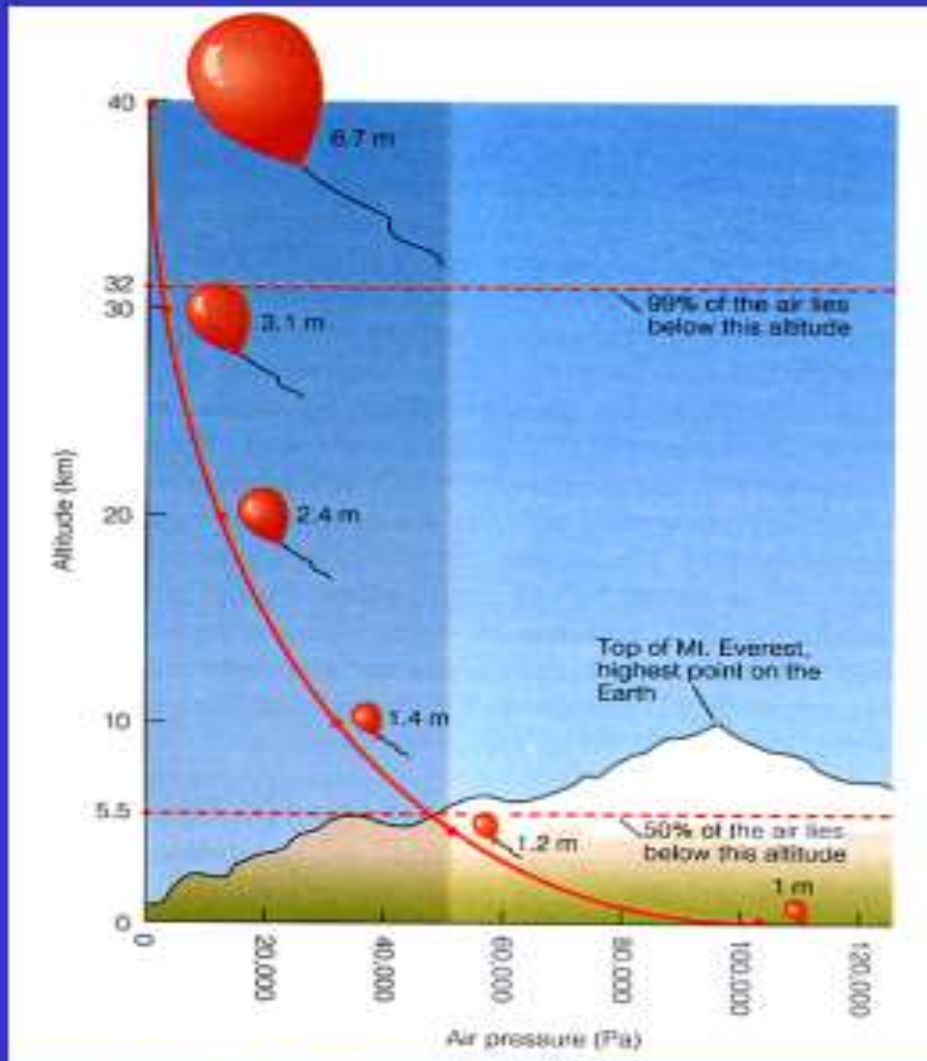
- Ascent leads to cooling $\rightarrow$ eventually reaches saturation.
- Further ascent increases RH $\rightarrow$ vapor pressure $>$ equilibrium.
- Excess vapor condenses $\rightarrow$ water droplets or ice form.

Adiabatic Process

- ❑ If a material changes its state (pressure, volume, or temperature) without any heat being added to it or withdrawn from it, the change is said to be adiabatic.
- ❑ The adiabatic process often occurs when air rises or descends and is an important process in the atmosphere.



Air Parcel Expands As It Rises...



(from *The Blue Planet*)

- ☐ Air pressure decreases with elevation.
- ☐ If a helium balloon 1 m in diameter is released at sea level, it expands as it floats upward because of the pressure decrease. The balloon would be 6.7 m in diameter as a height of 40 km.



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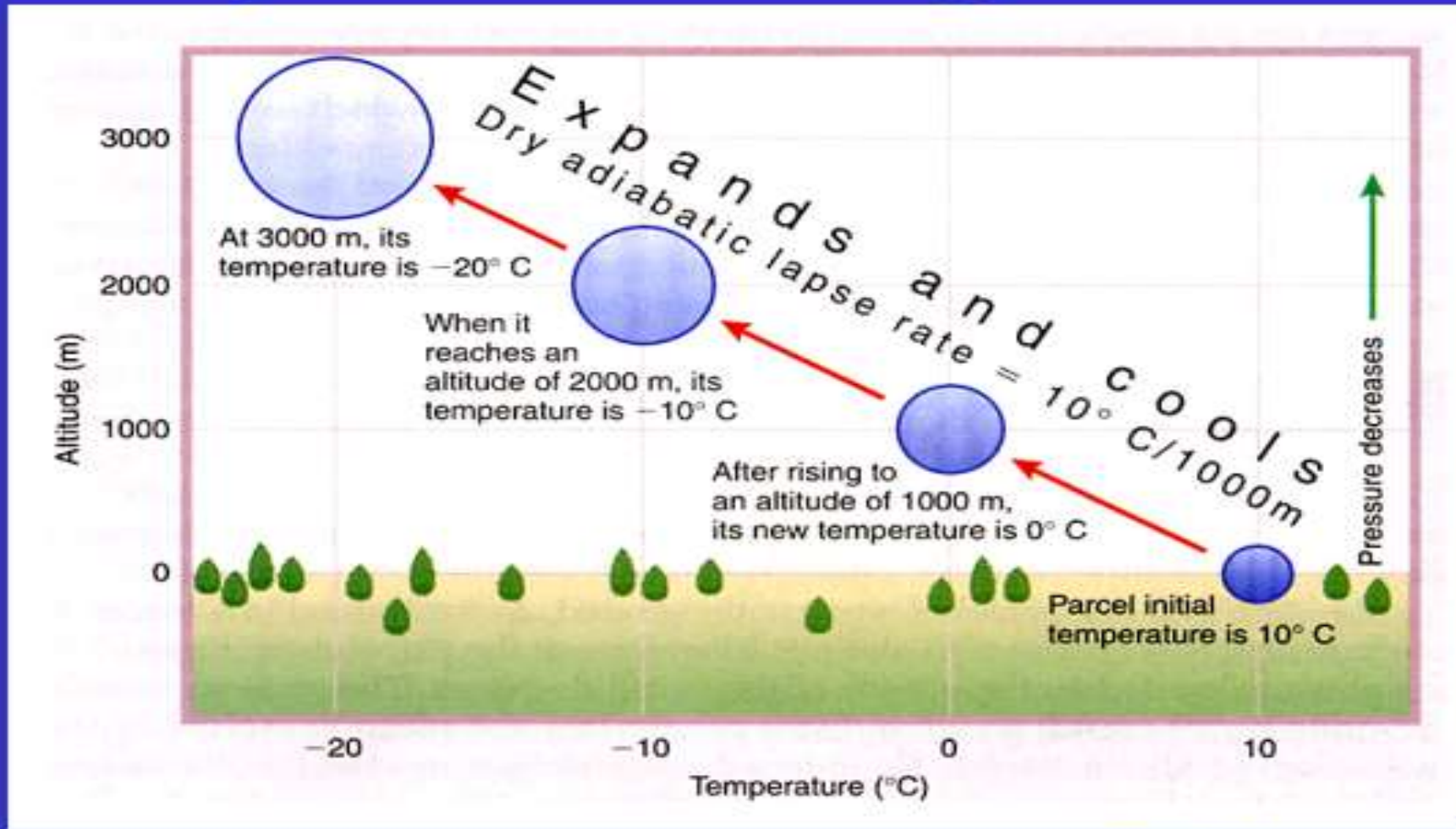
What Happens to the Temperature?

- ❑ Air molecules in the parcel (or the balloon) have to use their kinetic energy to expand the parcel/balloon.
- ❑ Therefore, the molecules lost energy and slow down their motions
 - ➔ The temperature of the air parcel (or balloon) decreases with elevation. The lost energy is used to increase the potential energy of air molecular.
- ❑ Similarly when the air parcel descends, the potential energy of air molecular is converted back to kinetic energy.
 - ➔ Air temperature rises.



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Dry Adiabatic Lapse Rate

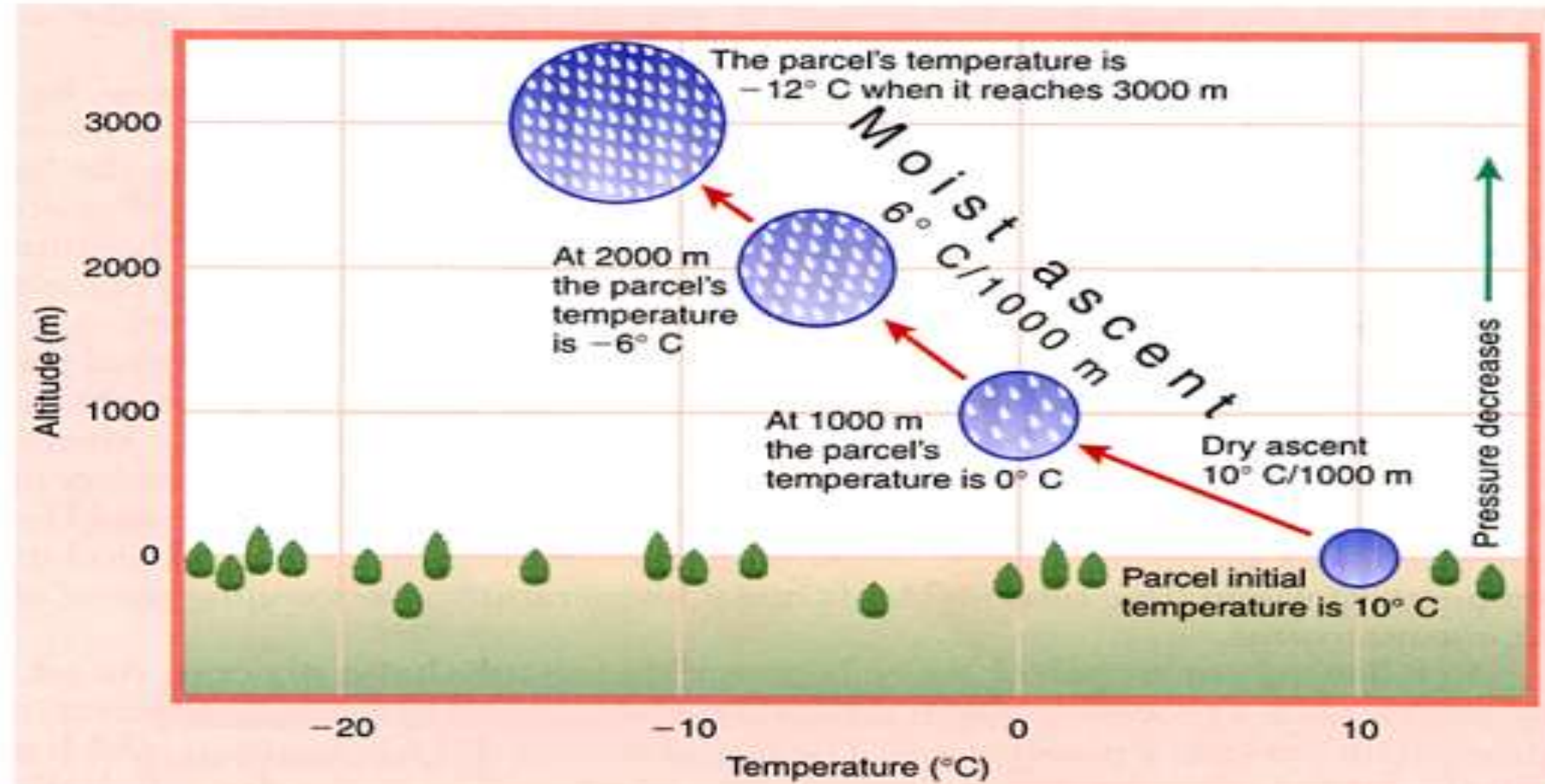


(from *Meteorology: Understanding the Atmosphere*)



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Moist Adiabatic Lapse Rate



(from *Meteorology: Understanding the Atmosphere*)



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Dry and Moist Adiabatic Lapse Rates

- ❑ Dry adiabatic lapse rate is constant = $10^{\circ}\text{C}/\text{km}$.
- ❑ Moist adiabatic lapse rate is NOT a constant. It depends on the temperature of saturated air parcel.
- ❑ The higher the air temperature, the smaller the moist adiabatic lapse rate.
- ➔ When warm, saturated air cools, it causes more condensation (and more latent heat release) than for cold, saturated air.



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Environmental Lapse Rate

- ❑ The environmental lapse rate is referred to as the rate at which the air temperature surrounding us would be changed if we were to climb upward into the atmosphere.
- ❑ This rate varies from time to time and from place to place.



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How Much Water Vapor Is Evaporated Into the Atmosphere Each Year?

- ❑ On average, 1 meter of water is evaporated from oceans to the atmosphere each year.
- ❑ The global averaged precipitation is also about 1 meter per year.

How Much Heat Is Brought Upward By Water Vapor?

- ❑ Earth's surface lost heat to the atmosphere when water is evaporated from oceans to the atmosphere.
- ❑ The evaporation of the 1m of water causes Earth's surface to lost 83 watts per square meter, almost half of the sunlight that reaches the surface.
- ❑ Without the evaporation process, the global surface temperature would be 67°C instead of the actual 15°C.

Exercise

1. Calculate the virtual temperature correction for moist air at 30 °C that has a mixing ratio of 20 g /kg
2. If air contains water vapor with a mixing ratio of 5.5 g /kg and the total pressure is 1026.8 hPa, calculate the vapor pressure e
3. Air at 1000 hPa and 18 °C has a mixing ratio of 6 g/ kg
What are the relative humidity and dew point of the air?

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Thanks !